

Chapter 20

Operational Risk

In 1999, bank regulators announced plans to assign capital for operational risk. This met with some opposition from banks. The chairman and CEO of one major international bank described it as “the dopest thing I have ever seen.” However, bank supervisors did not back down, because they knew that many of the big losses of banks over the previous 10 years had been operational risk losses—not credit risk or market risk losses.

The initial regulations gave banks a choice between two relatively simple ways of calculating operational risk capital (*the basic indicator approach* and the *standardized approach*) and one quite complicated approach (*the advanced measurement approach, AMA*). Regulators have since replaced these three approaches with a single “standardized measurement approach” that is simpler than AMA, but more complicated than the other two approaches. The new approach, which came into effect in 2023, is described in Section 20.4.

Operational risk has led to some big losses since the 1999 announcement (and this has probably helped to convince regulators that they were right to insist on requiring capital for operational risk). A big rogue trader loss at Société Générale in 2008 is discussed in Business Snapshot 5.5, and there was a similar loss at UBS in 2011. The London Whale loss at JPMorgan Chase in 2012 is discussed in Business Snapshot 22.1. In 2014, it was announced that the French bank BNP Paribas would pay \$9 billion (roughly one year’s profit) to the U.S. government for violating economic sanctions by moving dollar-denominated transactions through the American banking system on behalf of Sudanese, Iranian, and Cuban parties. The bank was also banned from conducting certain U.S. transactions for a year.

An increasingly important type of operational risk for banks is cyber risk. Banks have sophisticated systems in place to protect themselves from cyber attacks, but the attacks themselves are becoming more sophisticated. Also, banks are making increasing use of computer systems and the Internet, providing more avenues for cyber fraud. Customers and employees have to be continually educated so that the bank's data remain secure. Cyber attacks on banks are attractive to criminals because, to quote the bank robber Willie Sutton, "that's where the money is." They are also attractive to terrorists because of their potential to damage a nation's economic security and way of life.

Some regulators now regard operational risk as the most important risk facing banks. To quote Thomas J. Curry, head of the Office of the Comptroller of the Currency (OCC) in the United States, in 2012: "Given the complexity of today's banking markets and the sophistication of the technology that underpins it, it is no surprise that the OCC deems operational risk to be high and increasing. Indeed, it is currently at the top of the list of safety and soundness issues for the institutions we supervise." He goes on to argue that operational risk is more important than credit risk.¹ Most banks have always had some framework in place for managing operational risk. However, the prospect of new capital requirements led them to greatly increase the resources they devote to measuring and monitoring operational risk.

It is much more difficult to quantify operational risk than credit or market risk. Operational risk is also more difficult to manage. Financial institutions make a conscious decision to take a certain amount of credit and market risk, and there are many traded instruments that can be used to reduce these risks. Operational risk, by contrast, is a necessary part of doing business. An important part of operational risk management is identifying the types of risk that are being taken and should be insured against. There is always a danger that a huge loss will be incurred from taking an operational risk that ex ante was not even recognized as a risk.

It might be thought that a loss such as that at Société Générale (SocGen, see Business Snapshot 5.5) was a result of market risk because it was movements in market variables that led to it. However, it should be classified as an operational risk loss because it involved fraud. (Jérôme Kerviel created fictitious trades to hide the big bets he was making.) Suppose there was no fraud. If it was part of the bank's policy to let traders take huge risks, then the loss would be classified as a market risk loss. But, if this was not part of the bank's policy and there was a breakdown in its controls, it should be classified as an operational risk loss. The SocGen example illustrates that operational risk losses are often contingent on market movements. If the market had moved in Kerviel's favor, there would have been no loss. The fraud and the breakdown in SocGen's control systems might then never have come to light.

There are some parallels between the operational risk losses of banks and the losses of insurance companies. Insurance companies face a small probability of a large loss from a hurricane, earthquake, or other natural disaster. Similarly, banks face a small probability of a large operational risk loss. But there is one important difference. When insurance

¹Speech to the Exchequer Club, May 16, 2012.

companies lose a large amount of money because of a natural disaster, all companies in the industry tend to be affected and often premiums rise the next year to cover losses. Operational risk losses tend to affect only one bank. Because it operates in a competitive environment, the bank does not have the luxury of increasing prices for the services it offers during the following year.

20.1 Defining Operational Risk

There are many different ways in which operational risk can be defined. It is tempting to consider operational risk as a residual risk and define it as any risk faced by a financial institution that is not market risk or credit risk. To produce an estimate of operational risk, we could then look at the financial institution's financial statements and remove from the income statement (a) the impact of credit losses and (b) the profits or losses from market risk exposure. The variation in the resulting income would then be attributed to operational risk.

Most people agree that this definition of operational risk is too broad. It includes the risks associated with entering new markets, developing new products, economic factors, and so on. Another possible definition is that operational risk, as its name implies, is the risk arising from operations. This includes the risk of mistakes in processing transactions, making payments, and so on. This definition of risk is too narrow. It does not include major risks such as the risk of a rogue trader such as Jérôme Kerviel.

We can distinguish between internal risks and external risks. Internal risks are those over which the company has control. The company chooses whom it employs, what computer systems it develops, what controls are in place, and so on. Some people define operational risks as all internal risks. Operational risk then includes more than just the risk arising from operations. It includes risks arising from inadequate controls such as the rogue trader risk and the risks of other sorts of employee fraud.

Bank regulators favor including more than just internal risks in their definition of operational risk. They include the impact of external events such as natural disasters (for example, a fire or an earthquake that affects the bank's operations), political and regulatory risk (for example, being prevented from operating in a foreign country by that country's government), security breaches, and so on. All of this is reflected in the following definition of operational risk produced by the Basel Committee on Banking Supervision in 2001:

The risk of loss resulting from inadequate or failed internal processes, people, and systems or from external events.

Note that this definition includes legal risk but does not include reputation risk and the risk resulting from strategic decisions.

Operational risks result in increases in the bank's costs or decreases in its revenue. Some operational risks interact with credit and market risk. For example, when mistakes are made in a loan's documentation, it is usually the case that losses result if and only if

the counterparty defaults. When a trader exceeds limits and misreports positions, losses result if and only if the market moves against the trader.

20.2 Types of Operational Risk

The main categories of operational risk (as defined by regulators when they initially chose to require capital for operational risk) are:

1. *Internal fraud*: Acts of a type intended to defraud, misappropriate property, or circumvent regulations, the law, or company policy (excluding those concerned with diversity or discrimination) involving at least one internal party. Examples include intentional misreporting of positions, employee theft, insider trading on an employee's own account, and rogue trading.
2. *External fraud*: Acts by a third party of a type intended to defraud, misappropriate property, or circumvent the law. Examples include robbery, forgery, check kiting, and damage from computer hacking.
3. *Employment practices and workplace safety*: Acts inconsistent with employment, health, or safety laws or agreements, or that result in payment of personal injury claims or claims relating to diversity or discrimination issues. Examples include worker compensation claims, violation of employee health and safety rules, organized labor activities, discrimination claims, and general liability (for example, a customer slipping and falling at a branch office).
4. *Clients, products, and business practices*: Unintentional or negligent failure to meet a professional obligation to clients and the use of inappropriate products or business practices. Examples are fiduciary breaches, misuse of confidential customer information, improper trading activities on the bank's account, money laundering, and the sale of unauthorized products.
5. *Damage to physical assets*: Loss or damage to physical assets from natural disasters or other events. Examples include terrorism, vandalism, earthquakes, fires, and floods.
6. *Business disruption and system failures*: Disruption of business or system failures. Examples include hardware and software failures, telecommunication problems, and utility outages.
7. *Execution, delivery, and process management*: Failed transaction processing or process management, and disputes with trade counterparties and vendors. Examples include data entry errors, collateral management failures, incomplete legal documentation, unapproved access given to clients' accounts, non-client counterparty misperformance, and vendor disputes.

20.3 Loss Severity and Loss Frequency

Two distributions can be important in estimating the potential size of different categories of operational risk. One is the *loss frequency distribution* and the other is the *loss severity*

distribution. The loss frequency distribution is the distribution of the number of losses observed during one year. The loss severity distribution is the distribution of the size of a loss, given that a loss occurs.

For loss frequency, the natural probability distribution to use is a Poisson distribution. This distribution assumes that losses happen randomly through time so that in any short period of time Δt there is a probability $\lambda \Delta t$ of a loss occurring. The probability of n losses in T years is

$$e^{-\lambda T} \frac{(\lambda T)^n}{n!}$$

The parameter λ can be estimated as the average number of losses per year. For example, if during a 10-year period there were a total of 12 losses, λ would be estimated as 1.2 per year. A Poisson distribution has the property that the mean frequency of losses equals the variance of the frequency of losses.²

For the loss-severity probability distribution, a lognormal probability distribution is often used. The parameters of this probability distribution are the mean and standard deviation of the logarithm of the loss.

The loss-frequency distribution can be combined with the loss severity distribution for each risk type/business line combination to determine a loss distribution. Monte Carlo simulation can be used for this purpose.³ (The simplest assumption is that the two distributions are independent.) On each simulation trial, we proceed as follows:

1. We sample from the frequency distribution to determine the number of loss events ($= n$) in one year.
2. We sample n times from the loss severity distribution to determine the loss experienced for each loss event ($L_1, L_2, \dots, L_n$).
3. We determine the total loss experienced ($= L_1 + L_2 + \dots + L_n$).

When many simulation trials are used, we obtain a total loss distribution for losses of the type being considered. The 99.9 percentile of the distribution can then be determined.

Figure 20.1 illustrates the procedure. In this example, the expected loss frequency is 3 per year and the loss severity is drawn from a lognormal distribution. The logarithm of each loss (\$ millions) is assumed to have a mean of zero and a standard deviation of 0.4. The Excel worksheet used to produce Figure 20.1 is on the author's website: www-2.rotman.utoronto.ca/hull/riskman.

²If the mean frequency is greater than the variance of the frequency, a binomial distribution may be more appropriate. If the mean frequency is less than the variance, a negative binomial distribution (mixed Poisson distribution) may be more appropriate.

³Combining the loss severity and loss frequency distribution is a very common problem in insurance. Apart from Monte Carlo simulation, two approaches that are used are Panjer's algorithm and fast Fourier transforms. See H. H. Panjer, "Recursive Evaluation of a Family of Compound Distributions," *ASTIN Bulletin* 12 (1981): 22–29.

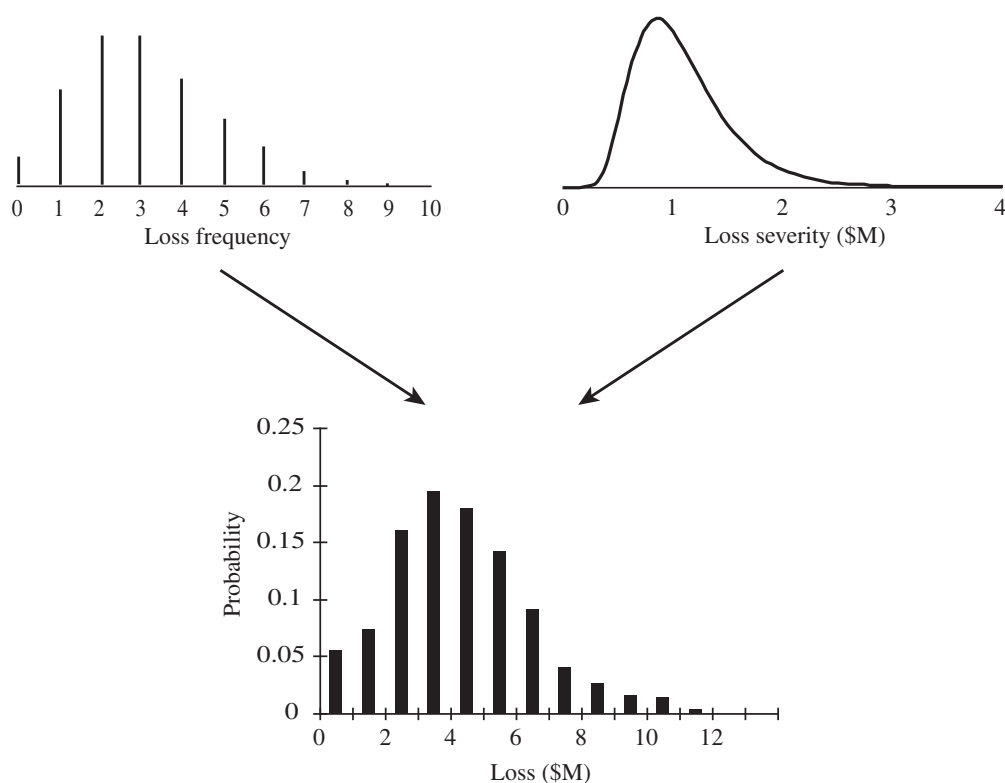


Figure 20.1 Calculation of Loss Distribution from Loss Frequency and Loss Severity

20.3.1 Internal vs. External Data

Unfortunately, there is usually relatively little historical data available within a bank to estimate loss severity and loss frequency distributions for particular types of losses. Many banks have not in the past kept records of operational risk losses. They are doing so now, but it may be some time before a reasonable amount of historical data is available. It is interesting to compare operational risk losses with credit risk losses in this respect. Traditionally, banks have done a much better job at documenting their credit risk losses than their operational risk losses. Also, in the case of credit risks, a bank can rely on a wealth of information published by credit rating agencies to assess probabilities of default and expected losses given default. Similar data on operational risk have not been collected in a systematic way.

Sometimes operational risk losses from other banks (external data) are used. There are two main ways in which this happens. The first involves data consortia, which are companies that facilitate the sharing of data between banks. (The insurance industry has had mechanisms for sharing loss data for many years and banks are now doing this as well.) The second is data vendors, who are in the business of collecting publicly available data in a systematic way. External data increases the amount of data available to a bank for estimating potential losses. It also has the advantage that it can lead to a consideration of

types of losses that have never been incurred by the bank, but which have been incurred by other banks.

Both internal and external historical data must be adjusted for inflation. In addition, a scale adjustment should be made to external data. If a bank with a revenue of \$10 billion reports a loss of \$8 million, how should the loss be scaled for a bank with a revenue of \$5 billion? A natural assumption is that a similar loss for a bank with a revenue of \$5 billion would be \$4 million. But this estimate is probably too small. For example, research by Shih et al. (2000) suggests that the effect of firm size on the size of a loss experienced is non-linear.⁴ Their estimate is

$$\text{Estimated Loss for Bank A} = \text{Observed Loss for Bank B} \times \left(\frac{\text{Bank A Revenue}}{\text{Bank B Revenue}} \right)^\alpha$$

where $\alpha = 0.23$. This means that in our example the bank with a revenue of \$5 billion would experience a loss of $8 \times 0.5^{0.23} = \$6.82$ million. After the appropriate scale adjustment, data obtained through sharing arrangements with other banks can be merged with the bank's own data to obtain a larger sample for determining the loss severity distribution.

The loss data available from data vendors are data taken from public sources such as newspapers and trade journals. Data from vendors cannot be used in the same way as internal data or data obtained through sharing arrangements because they are subject to biases. For example, only large losses are publicly reported, and the larger the loss, the more likely it is to be reported.

Data from vendors are most useful for determining *relative* loss severity. Suppose that a bank has good information on the mean and standard deviation of its loss severity distribution for internal fraud in corporate finance, but not for external fraud in corporate finance or for internal fraud in trading and sales. Suppose that the bank estimates the mean and standard deviation of its loss severity distribution for internal fraud in corporate finance as \$50 million and \$30 million. Suppose further that external data indicate that, for external fraud in corporate finance, the mean severity is twice that for internal fraud in corporate finance and the standard deviation of the severity is 1.5 times as great. In the absence of a better alternative, the bank might assume that its own severity for external fraud in corporate finance has a mean of $2 \times 50 = \$100$ million and a standard deviation of severity equal to $1.5 \times 30 = \$45$ million. Similarly, if the external data indicate that the mean severity for internal fraud in trading and sales is 2.5 times that for internal fraud in corporate finance and the standard deviation is twice as great, the bank might assume that its own severity for internal fraud in trading and sales has a mean of $2.5 \times 50 = \$125$ million and a standard deviation of $2 \times 30 = \$60$ million.

⁴See J. Shih, A. Samad-Khan, and P. Medapa, "Is the Size of an Operational Loss Related to Firm Size?" *Operational Risk Magazine* 2, no. 1 (January 2000). Whether Shih et al.'s results apply to legal risks is debatable. The size of a settlement in a large lawsuit against a bank can be governed by how much the bank can afford.

20.3.2 Scenario Analysis

Scenario analysis is a tool sometimes used for the assessment of operational risk. Some scenarios might come from the bank's own experience, some might be based on the experience of other banks, some might come from the work of consultants, and some might be generated by the risk management group in conjunction with senior management or business unit managers.

An operational risk committee consisting of members of the risk management group and senior management can be asked to estimate loss severity and loss frequency parameters for the scenarios. As explained earlier, a lognormal distribution is often used for loss severity and a Poisson distribution is often used for loss frequency. Data from other banks may be useful for estimating the loss severity parameters. The loss frequency parameters should reflect the controls in place at the bank and the type of business it is doing. They should reflect the views of the members of the operational risk committee. A number of categories of loss frequency can be defined, such as:

1. Scenario happens once every 1,000 years on average ($\lambda = 0.001$)
2. Scenario happens once every 100 years on average ($\lambda = 0.01$)
3. Scenario happens once every 50 years on average ($\lambda = 0.02$)
4. Scenario happens once every 10 years on average ($\lambda = 0.1$)
5. Scenario happens once every 5 years on average ($\lambda = 0.2$)

The committee can be asked to assign each scenario that is developed to one of the categories.

One difference between this scenario analysis and the one in Chapter 16 is that there is no model for determining losses and, if data are not available, the parameters of the loss severity distribution have to be estimated subjectively by the committee. One approach is to ask the committee to estimate an average loss and a high loss that the committee is 99% certain will not be exceeded. A lognormal distribution can then be fitted to the estimates.

Fortunately, the operational risk environment does not usually change as fast as the market and credit risk environment so that the amount of work involved in developing scenarios and keeping them up to date should not be as onerous. Nevertheless, the approach we have described does require a great deal of senior management time. The relevant scenarios for one bank are often similar to those for other banks and, to lessen the burden on the operational risk committee, there is the potential for standard scenarios to be developed by consultants or by bank industry associations. However, the loss frequency estimates should always be specific to a bank and reflect the controls in place in the bank and the type of business it is currently doing.

The advantage of generating scenarios using managerial judgment is that they include losses that the financial institution has never experienced, but could incur. The scenario analysis approach leads to management thinking actively and creatively about potential adverse events. This can have a number of benefits. In some cases, strategies for responding to an event so as to minimize its severity are likely to be developed. In other cases, proposals may be made for reducing the probability of the event occurring at all.

20.4 The Standardized Measurement Approach

In December 2017 the bank regulators finalized rules, referred to as the *standardized measurement approach* (SMA), for calculating capital for operational risk.⁵ The implementation date was January 2023.

Banks must first calculate what is known as the *business indicator* (BI). This is a measure of the bank's size and therefore its potential operational risk exposure. The BI has three components:

1. The interest leases and dividend component (ILDC)
2. The services component (SC)
3. The financial component (FC)

so that

$$BI = ILDC + SC + FC$$

The formulas for the components are:

$$ILDC = \text{Min}[\overline{[\text{Interest Income} - \text{Interest Expense}]}, 0.0225 \times \overline{\text{Interest Earning Assets}}] \\ + \overline{\text{Dividend Income}}$$

$$SC = \text{Max}[\overline{\text{Other Operating Income}}, \overline{\text{Other Operating Expense}}] \\ + \text{Max}[\overline{\text{Fee Income}}, \overline{\text{Fee Expense}}]$$

$$FC = |\overline{\text{Net P\&L Trading Book}}| + |\overline{\text{Net P\&L Banking Book}}|$$

with overbars indicating that averages are taken over the previous three years.

The next stage is to calculate the *business indicator component* (BIC) using the piecewise linear relationship in Table 20.1. The BIC is 12% of BI for small banks. Increases in BI between 1 and 30 billion euros lead to increases in BIC equal to 15% of the BI increase. Increases in BI above 30 billion euros lead to increases in BIC equal to 18% of the BI increase.

The *internal loss multiplier* (ILM) is then calculated as follows:

$$ILM = \ln \left[e - 1 + \left(\frac{LC}{BIC} \right)^{0.8} \right]$$

Table 20.1 Calculation of the BIC from the BI

BI (billions of euros)	BIC (billions of euros)
0 to 1	$0.12 \times BI$
1 to 30	$0.12 + 0.15 \times (BI - 1)$
>30	$4.47 + 0.18 \times (BI - 30)$

⁵See www.bis.org/bcbs/publ/d424.pdf.

BUSINESS SNAPSHOT 20.1

The Hammersmith and Fulham Story

Between 1987 and 1989, the London Borough of Hammersmith and Fulham in Great Britain entered into about 600 interest rate swaps and related transactions with a total notional principal of about six billion pounds. The transactions appear to have been entered into for speculative rather than hedging purposes. The two employees of Hammersmith and Fulham who were responsible for the trades had only a sketchy understanding of the risks they were taking and how the products they were trading worked.

By 1989, because of movements in sterling interest rates, Hammersmith and Fulham had lost several hundred million pounds on the swaps. To the banks on the other side of the transactions, the swaps were worth several hundred million pounds. The banks were concerned about credit risk. They had entered into offsetting swaps to hedge their interest rate risks. If Hammersmith and Fulham defaulted, they would still have to honor their obligations on the offsetting swaps and would take a huge loss.

What actually happened was not a default. Hammersmith and Fulham's auditor asked to have the transactions declared void because Hammersmith and Fulham did not have the authority to enter into the transactions. The British courts agreed. The case was appealed and went all the way to the House of Lords, Britain's highest court. The final decision was that Hammersmith and Fulham did not have the authority to enter into the swaps, but that they ought to have the authority to do so in the future for risk management purposes. Needless to say, banks were furious that their contracts were overturned in this way by the courts.

where the *loss component* (LC) is equal to 15 times average annual operational risk losses incurred over the previous 10 years. When high-quality data on losses over 10 years are not available, banks can use losses over five years during a transition period.

For small banks (BI less than one billion euros) the operational risk capital is set equal to BIC. For other banks it is set equal to the product of BIC and ILM. However, national bank supervisors have some discretion here. For example, they can set capital equal to BIC for all banks (while still requiring loss data disclosure). Alternatively, they can use loss data in determining capital for small banks.

To understand the formulas, note that when $LC = BIC$, $ILM=1$, and required capital equals BIC. Regulators expect that this will be true for an "average" bank. Banks experiencing operational risk losses that are higher than average its size will have an ILM

greater than 1, while those experiencing losses less than average will have an ILM less than 1.

20.5 Preventing Operational Risk Losses

A key part of an operational risk manager's job is to be proactive and try to prevent losses before they occur. We now consider some of the ways this can be done.

One approach is to monitor what is happening at other banks and try and learn from their mistakes. When a \$700 million rogue trader loss happened at a Baltimore subsidiary of Allied Irish Bank in 2002, risk managers throughout the world studied the situation carefully and asked: "Could this happen to us?" Business Snapshot 20.1 describes a situation concerning a British local authority in the late 1980s. It immediately led to all banks instituting procedures for checking that counterparties had the authority to enter into derivatives transactions.

20.5.1 Causal Relationships

Operational risk managers should try to establish causal relations between decisions taken and operational risk losses. Does increasing the average educational qualifications of employees reduce losses arising from mistakes in the way transactions are processed? Will a new computer system reduce the probabilities of losses from system failures? Are operational risk losses correlated with the employee turnover rate? Can the risk of a rogue trader be reduced by the way responsibilities are divided between different individuals and by the way traders are motivated?

One approach to establishing causal relationships is statistical. If we look at 12 different locations where a bank operates and find a high negative correlation between the education of back-office employees and the cost of mistakes in processing transactions, it might well make sense to do a cost-benefit analysis of changing the educational requirements for a back-office job in some of the locations. In some cases, a detailed analysis of the cause of losses may provide insights. For example, if 40% of computer failures can be attributed to the fact that the current hardware is several years old and less reliable than newer versions, a cost-benefit analysis of upgrading is likely to be useful.

20.5.2 RCSA and KRIs

Risk control and self-assessment (RCSA) is an important way in which banks try to achieve a better understanding of their operational risk exposures. It involves asking the managers of business units to identify their operational risks. Sometimes questionnaires and scorecards designed by senior management or consultants are used.

A by-product of any program to measure and understand operational risk is likely to be the development of key risk indicators (KRIs).⁶ Risk indicators are key tools in the

⁶These are sometimes referred to as Business Environment and Internal Control Factors (BEICFs).

management of operational risk. The most important indicators are prospective. They provide an early warning system to track the level of operational risk in the organization. Examples of key risk indicators that could be appropriate in particular situations are

1. Staff turnover
2. Number of failed transactions
3. Number of positions filled by temps
4. Ratio of supervisors to staff
5. Number of open positions
6. Percentage of staff who did not take 10 days consecutive leave in the last 12 months

The hope is that key risk indicators can identify potential problems and allow remedial action to be taken before losses are incurred. It is important for a bank to quantify operational risks, but it is even more important to take actions that control and manage those risks.

20.5.3 E-Mails and Phone Calls

One way in which operational risk costs can be mitigated is by educating employees to be very careful about what they write in e-mails and, when they work on the trading floor, what they say in phone calls. Lawsuits or regulatory investigations contending that a financial institution has behaved inappropriately or illegally are a major source of operational risk. One of the first things that happens when a case is filed is “discovery,” where the financial institution is required to provide all relevant internal communications. These have often proved to be very embarrassing and have made it difficult for financial institutions to defend themselves. (Fabrice Tourre’s e-mail in Business Snapshot 23.3 is an example of an embarrassing e-mail.) Before sending an e-mail or making a recorded phone call, an employee should consider the question “How could it hurt my employer if this became public knowledge?”

20.6 Allocation of Operational Risk Capital

Operational risk capital should be allocated to business units in a way that encourages them to improve their operational risk management. If a business unit can show that it has taken steps to reduce the frequency or severity of a particular risk, it should be allocated less capital. This will have the effect of improving the business unit’s return on capital (and possibly lead to the business unit manager receiving an increased bonus).

Note that it is not always optimal for a manager to reduce a particular operational risk. Sometimes the costs of reducing the risk outweigh the benefits of reduced capital so that return on allocated capital decreases. A business unit should be encouraged to make appropriate calculations and determine the amount of operational risk that maximizes return on capital.

The overall result of operational risk assessment and operational risk capital allocation should be that business units become more sensitive to the need for managing operational risk. Hopefully, operational risk management will be seen to be an important part of every manager's job. A key ingredient for the success of any operational risk program is the support of senior management. The Basel Committee on Banking Supervision is very much aware of this. It recommends that a bank's board of directors be involved in the approval of a risk management program and that it review the program on a regular basis.

20.7 Use of Power Law

In Section 8.4, we introduced the power law. This states that, for a wide range of variables

$$\text{Prob}(\nu > x) = Kx^{-\alpha}$$

where ν is the value of the variable, x is a relatively large value of ν , and K and α are constants. We covered the theoretical underpinnings of the power law and maximum likelihood estimation procedures when we looked at extreme value theory in Section 12.5.

De Fontnouvelle et al. (2003), using data on losses from vendors, found that the power law holds well for the large losses experienced by banks.⁷ This makes the calculation of VaR with high degrees of confidence, such as 99.9%, easier. Loss data (internal or external) and scenario analysis are employed to estimate the power law parameters using the maximum likelihood approach in Chapter 12. The 99.9 percentile of the loss distribution can then be estimated using equation (12.9).

When loss distributions are aggregated, the distribution with the heaviest tails tends to dominate. This means that the loss with the lowest α defines the extreme tails of the total loss distribution.⁸ Therefore, if all we are interested in is calculating the extreme tail of the total operational risk loss distribution, it may only be necessary to consider one or two business line/risk type combinations.

20.8 Insurance

An important decision for operational risk managers is the extent to which operational risks should be insured against. Insurance policies are available on many different kinds of risk ranging from fire losses to rogue trader losses. In Section 3.7 we discussed the moral hazard and adverse selection risks faced by insurance companies. We now review these again in the context of operational risk.

⁷See P. de Fontnouvelle, V. DeJesus-Rueff, J. Jordan, and E. Rosengren, "Capital and Risk: New Evidence on Implications of Large Operational Risk Losses," *Journal of Money, Credit and Banking* 38, no. 7 (October 2006): 1819–1846.

⁸The parameter ξ in extreme value theory (see Chapter 12) equals $1/\alpha$, so it is the loss distribution with the largest ξ that defines the extreme tails.

BUSINESS SNAPSHOT 20.2

Rogue Trader Insurance

A rogue trader insurance policy presents particularly tricky moral hazard problems. An unscrupulous bank could enter into an insurance contract to protect itself against losses from rogue trader risk and then choose to be lax in its implementation of trading limits. If a trader exceeds the trading limit and makes a large profit, the bank is better off than it would be otherwise. If a large loss results, a claim can be made under the rogue trader insurance policy. Deductibles, coinsurance provisions, and policy limits may mean that the amount recovered is less than the loss incurred by the trader. However, potential net losses to the bank are likely to be far less than potential profits, making the lax trading limits strategy a good bet for the bank.

Given this problem, it is perhaps surprising that some insurance companies do offer rogue trader insurance policies. These companies tend to specify carefully how trading limits are implemented within the bank. They may require that the existence of the insurance policy not be revealed to anyone on the trading floor. They are also likely to want to retain the right to investigate the circumstances underlying any loss.

From the bank's point of view, the lax trading limits strategy we have outlined may be very short-sighted. The bank might well find that the costs of all types of insurance rise significantly as a result of a rogue trader claim. Also, a large rogue trader loss (even if insured) would cause its reputation to suffer.

20.8.1 Moral Hazard

One of the risks facing a company that insures a bank against operational risk losses is moral hazard. This is the risk that the existence of the insurance contract will cause the bank to behave differently than it otherwise would. This changed behavior increases the risks to the insurance company. Consider, for example, a bank that insures itself against robberies. As a result of the insurance policy, it may be tempted to be lax in its implementation of security measures—making a robbery more likely than it would otherwise have been.

Insurance companies have traditionally dealt with moral hazard in a number of ways. Typically there is a *deductible* in any insurance policy. This means that the bank is responsible for bearing the first part of any loss. Sometimes there is a *coinsurance provision* in a policy. In this case, the insurance company pays a predetermined percentage (less than 100%) of losses in excess of the deductible. In addition, there is nearly always a *policy limit*. This is a limit on the total liability of the insurer. Consider again a bank that has

insured itself against robberies. The existence of deductibles, coinsurance provisions, and policy limits is likely to provide an incentive for a bank not to relax security measures in its branches. The moral hazard problem in rogue trader insurance is discussed in Business Snapshot 20.2.

20.8.2 Adverse Selection

The other major problem facing insurance companies is adverse selection. This is where an insurance company cannot distinguish between good and bad risks. It offers the same price to everyone and inadvertently attracts more of the bad risks. For example, banks without good internal controls are more likely to enter into rogue trader insurance contracts; banks without good internal controls are more likely to buy insurance policies to protect themselves against external fraud.

To overcome the adverse selection problem, an insurance company must try to understand the controls that exist within banks and the losses that have been experienced. As a result of its initial assessment of risks, it may not charge the same premium for the same contract to all banks. As time goes by, it gains more information about the bank's operational risk losses and may increase or reduce the premium charged. This is much the same as the approach adopted by insurance companies when they sell automobile insurance to a driver. At the outset, the insurance company obtains as much information on the driver as possible. As time goes by, it collects more information on the driver's risk (number of accidents, number of speeding tickets, etc.) and modifies the premium charged accordingly.

20.9 Sarbanes–Oxley

Largely as a result of the Enron bankruptcy, the Sarbanes–Oxley Act was passed in the United States in 2002. This provides another dimension to operational risk management for both financial and nonfinancial institutions in the United States. The act requires boards of directors to become much more involved with day-to-day operations. They must monitor internal controls to ensure risks are being assessed and handled well.

The act specifies rules concerning the composition of the board of directors of public companies and lists the responsibilities of the board. It gives the SEC the power to censure the board or give it additional responsibilities. A company's auditors are not allowed to carry out any significant non-auditing services for the company.⁹ Audit partners must be rotated. The audit committee of the board must be made aware of alternative accounting treatments. The CEO and CFO must prepare a statement to accompany the audit

⁹Enron's auditor, Arthur Andersen, provided a wide range of services in addition to auditing. The company did not survive the litigation that followed the downfall of Enron.

report to the effect that the financial statements are accurate. The CEO and CFO are required to return bonuses in the event that financial statements are restated. Other rules concern insider trading, disclosure, personal loans to executives, reporting of transactions by directors, and the monitoring of internal controls by directors.

Summary

In 1999, bank supervisors indicated their intention to charge capital for operational risk. This has led banks to carefully consider how they should measure and manage operational risk.

Part of the job of operational risk managers is to try to increase awareness among employees of operational risk so that losses are prevented. They should also try to understand what determines operational risk losses and develop key risk indicators to track the level of operational risk in different parts of the organization.

Once operational risk capital has been estimated, it is important to develop procedures for allocating it to business units. This should be done in a way that encourages business units to reduce operational risk when this can be done without incurring excessive costs.

The power law introduced in Chapter 8 seems to apply to operational risk losses. This makes it possible to use extreme value theory (see Chapter 12) to estimate the tails of a loss distribution from empirical data. When several loss distributions are aggregated, it is the loss distribution with the heaviest tail that dominates. In principle, this makes the calculation of VaR for total operational risk easier.

Many operational risks can be insured against. However, most policies include deductibles, coinsurance provisions, and policy limits. As a result, a bank is always left bearing part of any risk itself. Moreover, the way insurance premiums change as time passes is likely to depend on the claims made and other indicators that the insurance company has of how well operational risks are being managed.

The whole process of measuring, managing, and allocating operational risk is still in its infancy. As time goes by and data are accumulated, improvements to our procedures for managing operational risk are likely to emerge. In many ways, the key benefit of an operational risk management program lies in making managers more aware of the importance of operational risk and perhaps leading to them thinking about it differently.

Further Reading

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Practice Questions and Problems (Answers at End of Book)

- 20.1 What risks are included by regulators in their definition of operational risks? What risks are not included?
- 20.2 Suppose that external data show that a loss of \$100 million occurred at a bank with annual revenues of \$1 billion. Your bank has annual revenues of \$3 billion. What is the implication of the external data for losses that could occur at your bank? (Use Shih's result.)
- 20.3 Suppose that there is a 90% probability that operational risk losses of a certain type will not exceed \$20 million. The power law parameter, α , is 0.8. What is the probability of losses exceeding (a) \$40 million, (b) \$80 million, and (c) \$200 million?
- 20.4 Discuss how moral hazard and adverse selection are handled in car insurance.
- 20.5 Give two ways Sarbanes–Oxley affects the CEOs of public companies.
- 20.6 When is a trading loss classified as a market risk and when is it classified as an operational risk?
- 20.7 Discuss whether there is (a) moral hazard and (b) adverse selection in life insurance contracts.
- 20.8 What are external loss data? How are they obtained? How are they used in determining operational risk loss distributions for a bank?
- 20.9 What distributions are commonly used for loss frequency and loss severity?
- 20.10 Give two examples of key risk indicators that might be monitored by a central operational risk management group within a bank.

- 20.11 The worksheet used to produce Figure 20.1 is on the author's website. What is the mean and standard deviation of the loss distribution? Modify the inputs to the simulation to test the effect of changing the loss frequency from three to four.
- 20.12 What are the key inputs to determining regulatory capital under the Standardized Measurement Approach?
- 20.13 Consider two banks, each with a BIC of 200 million euros. Bank A has suffered 10 operational risk losses of 20 million euros in the past 10 years. Bank B has suffered one 200 million euro loss in the past 10 years. What is the operational risk regulatory capital for each bank under SMA?

Further Questions

- 20.14 Suppose that there is a 1% probability that operational risk losses of a certain type exceed \$10 million. Use the power law to estimate the 99.97% worst-case operational risk loss when the α parameter equals (a) 0.25, (b) 0.5, (c) 0.9, and (d) 1.0.
- 20.15 Consider the following two events: (a) A bank loses \$1 billion from an unexpected lawsuit relating to its transactions with a counterparty and (b) an insurance company loses \$1 billion because of an unexpected hurricane in Texas. Suppose that you have the same investment in shares issued by both the bank and the insurance company. Which loss are you more concerned about? Why?
- 20.16 The worksheet used to produce Figure 20.1 is on the author's website. How does the loss distribution change when the loss severity has a beta distribution with upper bound of five, lower bound of zero, and the other parameters both one?
- 20.17 A bank has a BI of 5.5 billion euros. It has had eight operational risk losses in the past 10 years. The amounts of the losses in millions of euros are: 3, 7, 15, 65, 85, 150, 250, and 300. What is the bank's operational risk regulatory capital under SMA?